Candidate Code – CC126



Subjects / Electives	Electronics Device and Circuits, FPGA base System Design, Analog CMOS Integrated Circuits, Analog Electronics, Digital Logic Design, VLSI Design, Computer Architecture, Modern Processor Architecture, Data Communication and Networking, Testing and Verification of Digital Circuits, Embedded System, Data Structures and Algorithms, Operation Systems, Object Originated Programming.
Technical Proficiency	**Languages & Skills:** Verilog, SystemVerilog, C, Python, ASIC, FPGA, RTL Design, Verification, DFT, ATPG, Communication Protocols **ASIC / Physical Design:** Synthesis, DFT, ATPG, Scan Insertion, Timing Analysis, Power Analysis, Area Analysis **Tools:** Cadence Genus / Modus / Xcelium, Xilinx Vivado, ModelSim, Quartus II, Keil uVision, Microwind, OpenLane, KLayout

SUMMER INTERNSHIP / WORK EXPERIENCE

Edge AI Fellow, Synseis (Semiconductor) — Jun 2026 - Present

- Developed a 64-32-16-10 MLP hardware architecture using Posit-16 (es=1) for digit recognition and evaluated it against an equivalent FP32 implementation for hardware-oriented inference.
- Analyzed FPGA PPA reports, reducing LUT utilization from 6,231 to 2,800 (55.1% reduction) while maintaining comparable estimated on-chip power of 0.331 W versus 0.334 W for FP32.
- Implemented and verified Posit arithmetic datapaths and MLP inference using Verilog/SystemVerilog and Python, including hardware-oriented weight representation.

ASIC Intern, eInfochips – An Arrow Company (Semiconductor) — May 2026 - Jun 2026

- Built an ESP32-based smart-home system with six sensors and an SSD1306 OLED; created a self-hosted Wi-Fi HTTP dashboard in Arduino IDE.
- Executed ASIC physical design of a D flip-flop with OpenLane and Sky130 PDK, achieving timing and power closure and producing a verified GDSII file in KLayout.

PROJECTS

RISC-V Processor with Custom AES Hardware Accelerator - Verilog, Xilinx Vivado

- Achieved single-cycle (1 clock cycle) latency for full AES-128 encryption, validated correct via the FIPS-197 known-answer test.
- Designed a custom RISC-V instruction-set extension with a fully unrolled 10-round hardware AES-128 datapath in Verilog HDL, verified through cycle-accurate simulation in Xilinx Vivado.

1×3 Network Packet Router & SystemVerilog Verification - SystemVerilog

- Achieved 266/266 data and 40/40 error checks by developing a self-checking SystemVerilog environment with generator, driver, monitor, scoreboard, mailboxes, and fault injection.
- Reached 100% signal and 96.67% transaction coverage by verifying destination, packet-length, and error scenarios across three output FIFOs.
- Fixed an RTL FIFO read defect and protocol-timing issues through cycle-accurate debugging and regression testing.

Scan-Based DFT and ATPG of Synchronous FIFO - Cadence Genus, Modus, Xcelium

- Achieved 99.98% adjusted stuck-at fault coverage (99.97% test coverage) on a synchronous FIFO design.
- Implemented a full scan-based DFT and ATPG flow with muxed scan insertion, automatic scan chain stitching, and stuck-at ATPG pattern generation using Cadence Genus and Modus.

Asynchronous FIFO Design and Verification - Verilog / SystemVerilog, Vivado

- Built a full verification environment for an asynchronous FIFO from the ground up, achieving 390/390 passing checks and 87.5–100% functional coverage across all covergroups.
- Implemented a layered SystemVerilog testbench with directed corner-case coverage and constrained-random stress testing, using separate generator, driver, monitor, and scoreboard layers with mailbox-based communication and virtual interfaces.

AXI4-Lite Master-Slave Communication System - Verilog / SystemVerilog, Xilinx Vivado

- Achieved 90% functional coverage by designing a protocol-compliant AXI4-Lite master-slave interface, implementing independent AW/W channels with zero deadlock conditions.
- Developed comprehensive SystemVerilog verification with an OOP testbench, assertions, and constrained-random testing.

AWARDS AND SCHOLARSHIPS

Received a 100% Tuition Fee Waiver (TFWS) Scholarship through merit-based selection for the top 5% based on Class 12 performance.

CERTIFICATIONS

CERTIFICATION	CERTIFYING AUTHORITY	DESCRIPTION
Foundations of VLSI Design Verification	Electronics & ICT Academies (MNIT Jaipur, IIITDM Jabalpur, IIT Roorkee & NIT Patna)	SystemVerilog, Verilog, RTL Design, Digital Design
System Design Through Verilog	IIT Guwahati	Aggregate: 81 / 100. Verilog, RTL Design, Digital System Design